

Week 3 — Q2 in Depth: From Population to Analytic Cohort

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES ·
previews L2

Week 3

Q2 in depth: from population to analytic cohort

A cohort is not a leftover. It is the first result.

- ▶ Week 2 asked *who is the population, and how is eligibility operationalised?* Today we answer it all the way down to executable code
- ▶ Everything on these slides — the funnel, the criteria table, the alcohol sentence — is the shape of what you build for **your own** paper at P1

Q2, in depth — and this week's division of labour

Week 2 gave you six questions. **Today is Q2:** who is the population, and how is eligibility operationalised?

	What it teaches	On which paper
Tuesday (today)	the judgment — turning a prose eligibility paragraph into logic you can defend	MASLD (Kueh)
Thursday (L2)	the mechanics — multi-cycle download, mortality linkage, eligibility steps, analytic dataset	benzodiazepine / opioid
P1	both, transferred	your locked paper

M0 — deadline today

- ▶ Memos due **12 PM today**
- ▶ We will get back to you soon on whether the teaching team agrees with your reading of the six criteria

Part 1 — The case definition

Target, source, analytic — and what each step costs

Population	Here	Why it narrows
Target	adults with MASLD in the US	who we want to speak about
Source	NHANES 2007–2018 respondents	who was sampled
Analytic	those surviving every filter, with measurable exposure <i>and</i> outcome	who we can actually compute on

- ▶ Each narrowing step trades **generalizability** for **measurability**
- ▶ A drop is not automatically a bias. **What it is depends on its kind** — Week 2's typed funnel: ELIGIBILITY, DESIGN, MISSINGNESS, TARGET POPULATION, OUTCOME ASCERTAINMENT
- ▶ Only one of those five is a missing-data problem

MASLD: a concept that has to become data operations

MASLD (formerly NAFLD) = **hepatic steatosis** + **metabolic dysfunction**, other causes not dominant.

Conceptual ingredient	What NHANES can offer	What the paper does
Hepatic steatosis	no biopsy, no imaging	Fatty Liver Index 60
Metabolic dysfunction	labs, exam, questionnaire	1 of five cardiometabolic criteria

- ▶ The constraint is real, not a flaw in this paper: every NHANES population study of MASLD this decade faces it
- ▶ **Nobody in this dataset was diagnosed with MASLD.** There is no column to read. Who counts as a case is the output of a calculation you run yourself — four measurements, one published formula, one threshold
- ▶ So who is in the cohort depends on how you handle these

Steatosis: the Fatty Liver Index, and where its inputs live

$$\text{FLI} = 100 \cdot \frac{\exp(y)}{1 + \exp(y)}, \quad y = 0.953 \ln(\text{TG}) + 0.139 \text{ BMI} + 0.718 \ln(\text{GGT})$$

Input	NHANES variable	Component	Fasting?
Triglycerides (mg/dL)	LBXTR	TRIGLY	yes
BMI (kg/m ²)	BMXBMI	BMX	no
GGT (U/L)	LBXSGTSI	BIOPRO	no
Waist (cm)	BMXWAIST	BMX	no

- ▶ Bedogni 2006. A published logistic score — **not** a product of the four
- ▶ **One** input needs a fasting lab, and that one input binds the whole index to the fasting subsample
- ▶ BMXBMI and BMXWAIST also build the *exposure*. Week 2 named that **structural dependence**; M1 dissects what it does to an estimate

FLI, worked: one subject

A 52-year-old with **TG 150 mg/dL**, **BMI 28.5**, **GGT 30 U/L**,
waist 99 cm:

Term	Computation	Value
$0.953 \ln(\text{TG})$	$0.953 \times \ln(150)$	4.775
0.139 BMI	0.139×28.5	3.962
$0.718 \ln(\text{GGT})$	$0.718 \times \ln(30)$	2.442
0.053 waist	0.053×99	5.247
constant		-15.745
	y	0.681

$$\text{FLI} = 100 \cdot \frac{\exp(0.681)}{1 + \exp(0.681)} = \mathbf{66.4} \Rightarrow \text{FLI} \geq 60 \Rightarrow \mathbf{\text{steatosis}}$$

- Note **TG** and **GGT** enter on the log scale — a doubling of GGT is worth the same as a doubling of TG times 0.75, not twice the raw units

The same subject, one lab value later

Change **only** the triglycerides — everything else identical:

	TG	y	FLI	Case?
Visit A	150 mg/dL	0.681	66.4	yes
Visit B	110 mg/dL	0.385	59.5	no

The cut flips at TG = 112.4 mg/dL, holding BMI, GGT and waist fixed.

- ▶ **40 mg/dL of triglycerides is the entire difference between “case” and “not a case”** — nothing else about these two rows differs
- ▶ So the one input that forced the cohort into the fasting subsample is also the one that decides case status at the margin
- ▶ **A cut on a continuous score** manufactures a hard boundary. The same issue returns at the **BMI × WHtR 2×2**

Metabolic dysfunction: five criteria, and 1 is required

#	Criterion	Operationalised as	Variables
1	Adiposity	BMI ≥ 25, or waist ≥ 94 cm (M) / ≥ 80 cm (F)	BMXBMI, BMXWAIST
2	Glycaemia	glucose ≥ 100, or HbA1c ≥ 5.7, or diabetes dx, or on treatment	LBXGLU, LBXGH, DIQ010, DIQ050, DIQ070
3	Blood pressure	SBP ≥ 130, or DBP ≥ 85, or on treatment	BPXSY1-4, BPXDI1-4, BPQ050A
4	Triglycerides	TG ≥ 150, or on lipid treatment	LBXTR, BPQ100D
5	HDL	40 mg/dL	LBDHDD

Part 2 — Eligibility as executable logic

Exclusions are scientific claims

- ▶ Every filter answers “*who is this study about?*” — so every filter needs a **stated rationale**, not just a line of code
- ▶ The reproducibility contract, in one rule:

```
# Paper, Methods: eligibility stated as "aged > 18 years"  
d <- filter(d, age >= 18)           # NOT what the sentence
```

- ▶ A comment naming **the sentence you are implementing** lets an auditor trace every `filter()` back to the paper
- ▶ It also makes a **divergence visible instead of invisible**. NHANES age is an integer, so `> 18` means 19 and over; we keep everyone aged exactly 18
- ▶ Week 2 put **three** age rules on one slide — the lab's `>= 20`, the paper's `> 18`, our `>= 18`. Same variable, three studies
- ▶ The discipline is not to pick the right one. It is to **record which one you used**, so a reader can undo it. A reproduction that quietly adopts the other rule has destroyed its own evidence

The sentence

The paper's Methods exclude significant alcohol consumption:

Three things it does not say:

	The question it leaves open
Which variable?	ALQ130 (drinks per <i>drinking day</i>) or $ALQ120Q \times ALQ120U$ (frequency $\times$ quantity)?
Per what?	per drinking day, or averaged over all days?
And the missing?	people who did not answer — dropped, or kept?

- ▶ ALQ130 is **drinks on days you drink**. Applying a “per day” rule to it directly is a different rule from the one the sentence appears to state

One sentence, three cohorts

Reading	Implemented as	Removes	Analytic N	vs the paper's 6,300
Crude — ALQ130 applied flatly	ALQ130 > 2 (M) / > 1 (F)	2,063	4,308	−1,992
Frequency-weighted — average drinks/day	quantity × frequency ÷ reference period	427	5,944	−356
No effective exclusion	—	0	6,371	+71

► A swing of **2,063 people** — a third of the cohort — from one

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The trap inside the trap: reserved codes

ALQ130 uses NHANES **reserved codes**, not measurements:

Code	Means
777	refused
999	don't know

```
qty <- ifelse(ALQ130 %in% c(777, 999), NA, ALQ130) # str
```

- ▶ Left in, ALQ130 > 2 silently reclassifies every refusal as a heavy drinker
- ▶ Here that is **nine people**: the crude exclusion moves **2,063** → **2,072**
- ▶ Two traps live on one variable and they are **two hundred times apart in size**. Do not fuse them in your notes
A reserved code is not a quantity. Strip them before every threshold rule, in every increment.

What we did, and why we are telling you

Our choice: no effective alcohol exclusion. It reproduces the published cohort and is the most transparent about the discrepancy.

```
## 02_build_analytic.R:40  
APPLY_ALCOHOL_EXCL <- FALSE
```

- ▶ It is a **switch**, not a hidden default: the funnel prints Alcohol exclusion applied = FALSE as a row, so the setting is in the artifact
- ▶ We carry the alcohol question forward as an **open finding**, not a solved step
- ▶ The discipline, stated once: **choose a priori, record the choice where a reader will find it, and report what the alternatives would have given**
We are not telling you which reading is right. We are showing you what it costs to pretend the question did not arise.

BREAK

On the board for the restart: *which of the three alcohol codings would you defend, and what would you write in the paper?*

Part 3 — The funnel

The funnel, with a type on every row

Step	Criterion	N	Dropped	Kind
0	NHANES 2007–2018 records	59,842	—	—
1	Age 18	36,580	23,262	ELIGIBILITY
2	In fasting subsample	14,962	21,618	DESIGN
3	FLI com- putable	14,170	792	MISSINGNESS
4	Steatosis (FLI 60)	6,052	8,118	TARGET POPULATION
5	MASLD (1 criterion)	6,050	2	TARGET POPULATION
6	Phenotype group assignable	6,050	0	MISSINGNESS
7	Mortality- eligible and	6,048	2	OUTCOME ASCERTAINMENT

Two row sets, and they are not interchangeable

	Locked domain	Full analytic file
Fasting subsample enforced as its own step	yes (row 2)	fused into “FLI computable”
Final N	6,048	6,371
What it is for	every design-aware analysis; carries WTSAF2YR	matching the paper’s reported cohort
Phenotype groups	—	391 / 4,585 / 787 / 608
Deaths	—	586

- ▶ The paper reports **6,300**, groups **386 / 4,541 / 769 / 604, 585 deaths** — those are *theirs*, and ours are the row above
- ▶ Quoting an N without naming its row set is the error this course exists to catch
Two numbers that differ by 323 people are not a discrepancy if they are answers to different questions. They are

What the funnel reveals

- ▶ The single largest drop is **age 18** — 23,262 people. That is the study's own eligibility rule working as intended
- ▶ **After eligibility**, the two largest drops are the **fasting subsample** (21,618, DESIGN) and the **disease definition** (8,118, TARGET POPULATION)
- ▶ **Neither of those is missing data.** The genuine missingness in this table is $792 + 0$
- ▶ Within the fasting subsample, FLI is computable for **94.7%** of adults
Feasibility, before you commit: is the exposure computable for enough of your target population? Here the answer arrives as a design fact, not a data-quality problem.

Why you never tune the cohort to hit the paper's N

- ▶ The temptation: quietly flip a threshold until N matches the abstract
- ▶ That is **fitting the process to the answer** — you have chosen your method using the result, and the method no longer carries evidence
- ▶ One sentence in this paper moves N by **2,063**. So an exact match is roughly as likely to mean *you guessed their ambiguity correctly* as *you implemented it correctly*

	What it demonstrates
Matched N, undocumented coding	nothing — the match is unattributable
Mismatched N, traced to a named step	the reproduction worked

A documented N you can defend beats a matched N you cannot.

Reviewing someone else's eligibility

Ask	On this paper
Is the case definition operational ?	mostly — FLI 60 is exact; “ 1 criterion” needs the five spelled out
Is every exclusion executable as written?	no — the alcohol criterion is not
Does the funnel's arithmetic close ?	not reported; we had to rebuild it
Is the analytic N stated and reproducible ?	stated; reproducible only up to the above

- ▶ Also check the tables against the Methods: here the Table 1–2 headers read “**BMI < 25**” while the Methods define obesity at **30**, and the non-obese groups have BMI medians of **27.6** and **28.8** — so the headers appear to be a labelling error
- ▶ **Charity and rigour together:** assume competence, and still require that the numbers reconcile

Part 4 — Thursday

L2 assembles the same object from a different paper

EpiMethods accessingE2 — **benzodiazepine / opioid use and all-cause mortality**, NHANES 1999–2014.

```
dat.mort    <- rbind(dat.mortality99, dat.mortality01, ...,
dat.nhanes  <- rbind(dat.nhanes99,      dat.nhanes01, ...,

dat <- merge(dat.mort, dat.nhanes, by = "SEQN", all = TRUE)
```

- ▶ **Eight cycles stacked, then merged on SEQN** — the same key and the same pattern L1 used with `full_join(demo, bmx, by = "SEQN")`
- ▶ `all = TRUE` is the `full_join` decision again, made with a different word. Week 2's rule stands: **the join verb defines the analysis set**
- ▶ Mortality is a **separate linked file** here too — it is not an NHANES component

And L2 writes a funnel — it just does not know it yet

```
dat1 <- subset(dat, RIDAGEYR >= 20)
dat2 <- subset(dat1, RIDSTATR == "int + mec exam")
dat3 <- subset(dat2, TAKERX_INLASTMONTH %in% c("No", "Yes"))
dat4 <- dat3[complete.cases(dat3$BPQ020), ]
dat5 <- subset(dat4, mortstat == 0 | (mortstat == 1 & permstat == 1))
dat6 <- subset(dat5, benzodiazepines == "Yes" | ssris == "Yes")
```

- ▶ Six eligibility steps, **N after each one** — that is a funnel, written as comments
- ▶ Now type it, the way we typed ours:

	dat1	dat2	dat3	dat4	dat5	dat6
kind	ELIGIBILITY	DESIGN	MISSINGNESS	MISSINGNESS	ELIGIBILITY	TARGET POPULATION

- ▶ **dat4 prints no N.** The complete-case steps are the ones that go uncounted — in this lab and in most papers

One more thing to notice on Thursday

L2's exposure has four levels, and the reference category is **"Neither"**:

```
dat6 <- subset(dat5, benzodiazepines == "Yes" | ssris == "Yes")
dat6$exposure <- relevel(dat6$exposure, ref = "Neither")
```

- ▶ Eligibility required **at least one** of three drug classes. So "Neither" means *neither benzodiazepines nor opioids* — **everyone in it is an SSRI user**
- ▶ The reference group is an **active comparator**, not an unexposed population
- ▶ Exactly the shape of our Group I: every hazard ratio in the MASLD paper is measured against a group that is **obese**
A ratio without its referent means nothing — and in both studies the referent is not "healthy".

P1 — the same two things, on your paper

- ▶ **Rebuild the cohort** from raw source files: list every file and cycle, name the merge key, write each inclusion and exclusion as **executable logic with a comment citing the sentence**
- ▶ **Produce the funnel**: N removed and N remaining at every step, typed
- ▶ **Flag one ambiguous eligibility sentence** you had to adjudicate — and say what the alternatives would have given
- ▶ Look variables up in the **NHANES variable map** rather than guessing: it says which cycles a variable exists in, and whether 77 is a code or a measurement
- ▶ **Do not aim at a number.** The paper reports 6,300; our full analytic file lands on 6,371; yours is wherever your documented steps take you
- ▶ P1 is **optional and ungraded** — no deadline, nothing to submit. Keep the script, the flow table and P1_notes.md in your group repo. It is assembled into **M1**
Be ready for: “show me one person who is in your cohort but not the paper’s — and tell me why.”

Where we landed

- ▶ Q2 answered all the way down: a **conceptual** case definition became an **operational** one, and the operations turned out to be **contestable**
- ▶ A cut on a continuous score decides case status at the margin; a five-criterion definition hides five threshold choices; one prose sentence yields **three cohorts**
- ▶ The funnel with a type on every row is what makes any of it checkable
- ▶ **Thursday:** the same assembly, a different paper. If you can only do it on MASLD, you have not learned it

Reading a Methods section is the first act of reproduction.

Building the cohort is the second.

Reference

Kueh MTW, Intaran MAU, Goh R, et al. **Body weight categories and fat distribution in relation to all-cause mortality among adults with metabolic dysfunction-associated steatotic liver disease: a population-based analysis of NHANES, 2007–2018.** *BMJ Open*. 2026;16:e113719. doi: 10.1136/bmjopen-2025-113719

Bedogni G, Bellentani S, Miglioli L, et al. **The Fatty Liver Index: a simple and accurate predictor of hepatic steatosis in the general population.** *BMC Gastroenterol*. 2006;6:33.

Lab exercise: EpiMethods, Accessing & preparing NHANES–mortality data.

Cohort counts, funnel types and the three alcohol codings on these slides come from this repository's own reproduction pipeline (`reproduction/logs/sample_funnel.csv`, `examples/_results/P1.json`), not from the paper's prose. The worked FLI values are computed from the Bedogni formula as implemented at `02_build_analytic.R`:140